

Rocket flight

Y16 (2016) — English translation

A space rocket rotates in a circular orbit at a distance $R = 2.0 \cdot 10^7$ m from the center of the Earth in the plane of the Earth's orbit. Earth mass $m = 6.00 \cdot 10^{24}$ kg, Sun mass $M = 2.0 \cdot 10^{30}$ kg, Earth orbital radius $R_0 = 1.5 \cdot 10^{11}$ m, gravitational constant $G = 6.67 \cdot 10^{-11} (\text{N} \cdot \text{m}^2)/\text{kg}^2$.

A1 Find the minimum value of the change in speed Δv of the rocket, which may be sufficient for its departure from orbit outside the Solar System. 5.00
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Source: <https://pho.rs/p/3031>